

Virtual Servers:

A virtual server is a logical entity created by FortiADC that represents the frontend of a service. Clients connect to the virtual server, and FortiADC then directs the traffic to the appropriate real server from the server pool.

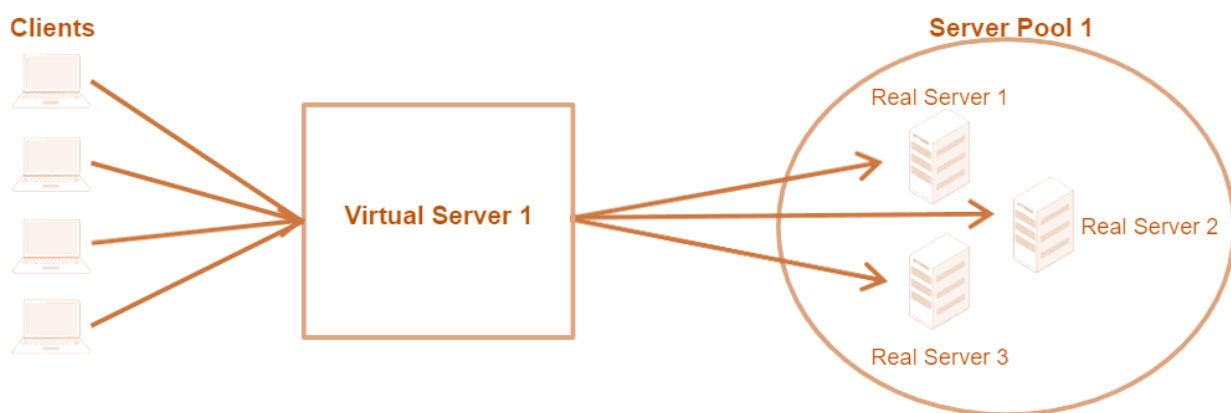
A virtual server could be the IP address or domain name that clients use to connect to the application. The virtual server acts as the entry point for client requests.

Virtual Server acts as an intermediary between clients and the actual servers hosting the applications. It receives incoming traffic and distributes it across multiple backend servers to optimize resource use, maximize throughput, and ensure high availability. Listens for incoming traffic on specific ports and IP addresses and then directs that traffic based on predefined rules.

A **Virtual Server** sits between users and actual servers to manage traffic efficiently. Users think they are connecting to a single server, but the virtual server distributes their requests to multiple real servers in the background. This setup improves security, speeds up performance, and balances the load across servers.

Server pools are assigned to virtual servers, and each server pool contains one or more real servers.

Real servers are the actual machines or applications that handle user requests. They run services like HTTP, FTP, FTPS, RADIUS, and others, processing data and delivering content to users. The virtual server directs traffic to these real servers based on load balancing and security policies.



Layer 7 ADC:

Layer 7 Virtual Servers operate at the Application Layer, making decisions based on HTTP headers, cookies, and other application-level data. This is particularly useful for applications that require more intelligent distribution of traffic, such as web applications. They can analyze and modify HTTP traffic, such as rewriting headers or filtering content. IPv4 and IPv6 are supported on layer 7 virtual servers. This makes Layer 7 Virtual Servers more intelligent and secure compared to Layer 4.

Key Features:

1. Match Rules

- Define routing rules based on content.
- Examples: File type (e.g., text vs. graphics).

2. Load Balancing Methods

- Round Robin
- Weighted Round Robin
- Least Connections
- Fastest Response

3. Health Checks

- Layer 3, 4, or 7 data-based checks.
- Ensures backend server availability.

4. Persistence

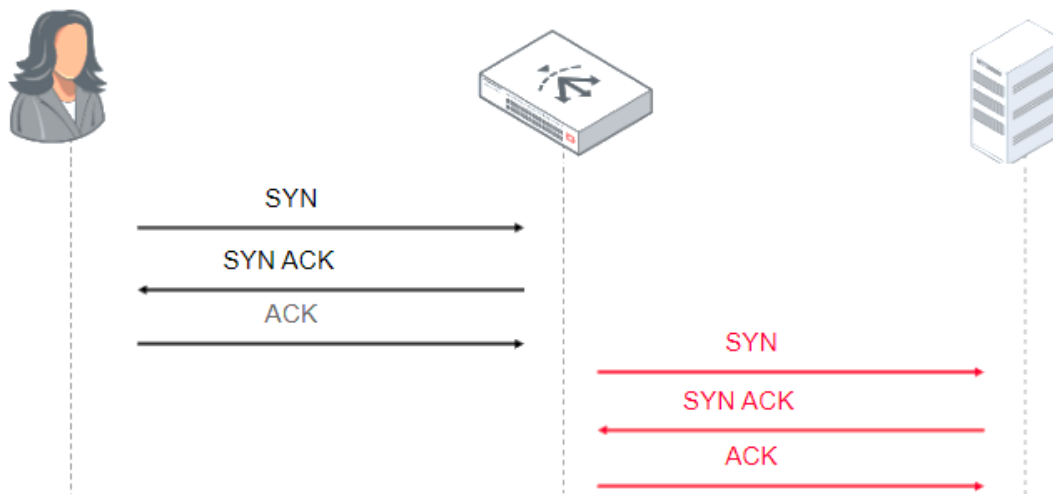
- Based on cookies, TCP/IP headers, TLS/SSL session IDs, etc.

5. Content Routing

- Route traffic based on HTTP Host, Referrer, Request URL, etc.

6. Content Rewriting

- URL rewriting.
- HTTP request/response header modifications.

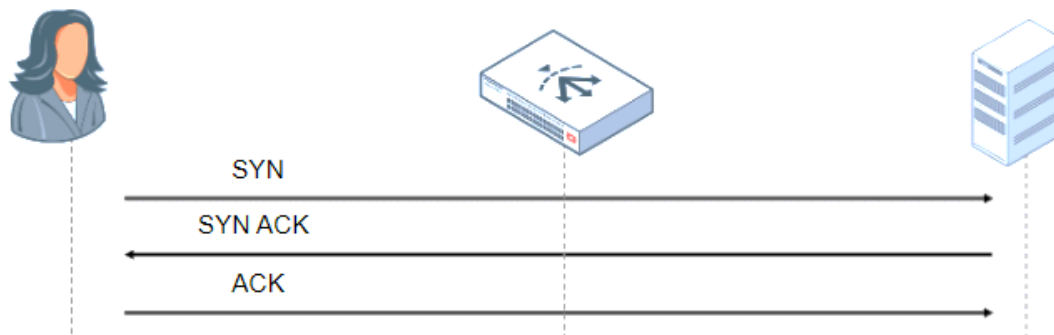


Layer 4 ADC:

Layer 4 Virtual Servers process traffic based on the first packet of a new session, making them the fastest option. They support both IPv4 and IPv6. By default, Destination NAT (DNAT) is used to forward packets to real servers. However, if FortiADC is not the default gateway, it may cause asymmetric routing issues, where request and response packets take different paths, potentially disrupting communication. It routes network traffic based on information from the TCP and UDP protocols without inspecting the actual content of the traffic.

Key Features:

1. **Protocol-Based Routing**
 - Routes traffic based on TCP/UDP port no, providing efficient distribution of data.
2. **Load Balancing Methods**
 - **Round Robin:** Distributes requests evenly across all servers.
 - **Weighted Round Robin:** Takes server capacity into account.
 - **Least Connections:** Directs traffic to the server with the fewest active connect.
 - **Fastest Response:** Routes to the server with the quickest response time.
3. **Health Checks**
 - **Layer 3 and 4 Monitoring:** Ensures the availability of backend servers by monitoring their health and performance.
 - Methods include ping, TCP/UDP, and custom script health checks.
4. **Performance and Scalability**
 - **High Throughput:** Can handle a large volume of connections with minimal latency.
 - **Scalability:** Supports scaling out as traffic demands grow, ideal for high-performance applications.
5. **Connection Persistence**
 - **Session Persistence:** Ensures continuous connection by keeping the client connected to the same server, using source IP address or other criteria.
6. **Security Features**
 - **Firewall Integration:** Works with firewall policies for enhanced security.
 - **DDoS Protection:** Mitigates Distributed Denial of Service attacks at Layer 4.



Layer 2 ADC:

Layer 2 Virtual Servers in FortiADC distribute traffic among multiple next-hop gateways, similar to Layer 7 but at the data link layer. Layer 2 load balancing is useful when the request's destination IP is unknown, and you need to balance connections between multiple next-hop gateways. Supports HTTP content inspection and modification.

Introduction

- Layer 2 (L2) load balancing allows traffic distribution without requiring changes to the network infrastructure.
- Used primarily in environments where direct traffic manipulation is necessary without altering IP addressing schemes.

Benefits

- **Simplicity:** Easy to implement as it operates at the data link layer.
- **Efficiency:** Reduces latency by directing traffic based on MAC addresses.
- **Compatibility:** Works seamlessly with existing network configurations and devices.

How It Works

1. **Traffic Ingress:** Incoming traffic enters the network through an L2 switch.
2. **MAC Address Assignment:** FortiADC assigns & balances traffic based on MAC addresses.
3. **Traffic Distribution:** Traffic is distributed to the appropriate servers without modifying the IP headers.

Key Features

- **Transparency:** Operates without altering packet headers.
- **Scalability:** Easily scales with network growth.
- **Fault Tolerance:** Ensures high availability and reliability.

Use Cases

- Data centers requiring efficient load distribution.
- Environments with non-routable IP address configurations.
- Networks needing minimal configuration changes.

